PROBLEM CL12

Gas Discharging through an Inverted U-Pipe: Outlet State, Force and Moment

Original statement · CL12.pdf

Compressible | Energy | Linear momentum | Angular momentum

OFFICIAL PROBLEM STATEMENT

The gas circulating along the pipe of cross section A depicted in the figure of CL12.pdf discharges to the atmosphere, where the pressure is p_a. Measurements at a given upstream location give V_u, p_u, and T_u for the velocity, pressure and temperature, all uniform across the section. In the analysis that follows, it is assumed that the pipe is thermally insulated and that buoyancy forces have a negligible effect on the gas motion. The gas properties are defined by its gas constant R_g together with the ratio of specific heats $\gamma = c_p/c_v$.

- Obtain the values of the gas temperature and velocity at the outlet section T_o and V_o in terms of V_u, p_u, T_u, p_a, R_g and γ.
- Compute the two components of the force F_x and F_y acting on the stretch of pipe considered.
- Calculate the momentum on the pipe with respect to the origin of the coordinate system indicated in the figure.

Source: CL12.pdf (page 1, exam of Dec 16, 2011). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady compressible flow of a perfect gas ($p = \rho R_g T$, $h = c_p T = \frac{\gamma}{\gamma-1} R_g T$); neglect $\bar{g}$; pipe insulated and at rest (no heat, no work); uniform properties at both sections.
- Geometry (figure): inverted U of constant section A; the gas goes up the left leg ($\bar{v} = V_u \bar{e}_y$ at the measuring section) and down the right leg, discharging as a free jet ($p = p_a$, $\bar{v} = -V_o \bar{e}_y$) at a horizontal distance L from the left leg; the origin O is on the axis of the left leg.

- Data: $[V_u,\ p_u,\ T_u,\ p_a,\ R_g,\ \gamma,\ A,\ L]$. Unknowns: $[T_o,\ V_o,\ F_x,\ F_y,\ \bar{M}]$. The 'momentum on the pipe' of the statement is the moment of the forces on the pipe about O.

Symbol	Meaning	SI unit
V_u, p_u, T_u	upstream state	m/s, Pa, K
V_o, T_o	outlet velocity and temperature	m/s, K
A, L	pipe section, distance between legs	m^2, m
$\bar{F}, \bar{M}$	force and moment on the pipe	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the pipe; $\bar{e}_x$ horizontal (from the left leg to the right leg), $\bar{e}_y$ vertical upwards, $\bar{e}_z = \bar{e}_x \wedge \bar{e}_y$ towards the reader. Fixed control volume.
- Surfaces: Σ_u (measuring section, $\bar{n} = -\bar{e}_y$), Σ_o (outlet, $\bar{n} = -\bar{e}_y$), Σ_w (inner wall). Positions: $\bar{x} = y\bar{e}_y + \bar{\eta}$ on Σ_u and $\bar{x} = L\bar{e}_x + y\bar{e}_y + \bar{\eta}$ on Σ_o, with $\int \bar{\eta}\, d\sigma = 0$.

$$\boxed{\bar{F} = \int_{\Sigma_w} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_u	$-\rho_u V_u A$, with $\rho_u = \frac{p_u}{R_g T_u}$
Σ_o	$+\rho_o V_o A$, with $\rho_o = \frac{p_a}{R_g T_o}$
Σ_w	0

$$\rho_u V_u A = \rho_o V_o A \implies \boxed{\frac{p_u V_u}{T_u} = \frac{p_a V_o}{T_o}} \qquad (1)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\left(e+\frac{v^2}{2}\right)dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)\left[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right]d\sigma=-\int_{\Sigma_c}p\,\bar{v}\cdot\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}}_{\text{neglect }\bar{g}}-\int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}Q_H\,dV}}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_u	$-\dot{m}\left(e_u+\frac{V_u^2}{2}\right)$	$+p_u V_u A$	0	0
Σ_o	$+\dot{m}\left(e_o+\frac{V_o^2}{2}\right)$	$-p_a V_o A$	0	0
Σ_w	0	0 ($\bar{v}=0$)	0 ($\bar{v}=0$)	0 (insulated)

$$\boxed{\frac{\gamma}{\gamma-1}R_gT_u+\frac{1}{2}V_u^2=\frac{\gamma}{\gamma-1}R_gT_o+\frac{1}{2}V_o^2} \tag{2}$$

Solving (1) and (2): with $C\equiv\frac{p_uV_u}{p_aT_u}$, (1) gives $V_o=C\,T_o$; substituting into (2), $\frac{C^2}{2}T_o^2+\frac{\gamma R_g}{\gamma-1}T_o-H=0$ with $H=\frac{\gamma R_g}{\gamma-1}T_u+\frac{V_u^2}{2}$, whose positive root is

$$\boxed{T_o=\frac{-\frac{\gamma R_g}{\gamma-1}+\sqrt{\left(\frac{\gamma R_g}{\gamma-1}\right)^2+2C^2H}}{C^2},\qquad V_o=C\,T_o} \tag{3}$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_u	$\rho_uV_u\bar{e}_y(-V_u)A=-\rho_uV_u^2A\,\bar{e}_y$	$-(p_u-p_a)(-\bar{e}_y)A=(p_u-p_a)A\,\bar{e}_y$	0
Σ_o	$\rho_o(-V_o\bar{e}_y)(V_o)A=-\rho_uV_uA\,V_o\,\bar{e}_y$	0 (free jet)	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}$. The Σ_o entry uses (1).

$$-\rho_uV_uA\,(V_u+V_o)\,\bar{e}_y=(p_u-p_a)A\,\bar{e}_y-\bar{F}\implies\boxed{F_x=0,\qquad F_y=\rho_uV_uA\,(V_u+V_o)+(p_u-p_a)A} \tag{4}$$

Angular momentum about O (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,\bar{x}\wedge\bar{v}\,dV}}_{0} + \int_{\Sigma_c} \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_u	0 ($y\bar{e}_y\wedge\bar{e}_y=0$, $\int\bar{\eta}\,d\sigma=0$)	0 (same reason)	0
Σ_o	$\rho_o V_o A\,(L\bar{e}_x\wedge(-V_o\bar{e}_y)) = -\rho_u V_u A\,LV_o\,\bar{e}_z$	0 (free jet)	0
Σ_w	0	$-\int_{\Sigma_w}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$

Boxed Σ_w terms: $-\bar{M}$.

$$-\rho_u V_u A\,LV_o\,\bar{e}_z = -\bar{M} \implies \boxed{\bar{M} = \rho_u V_u A\,L\,V_o\,\bar{e}_z = \frac{p_u}{R_g T_u} V_u A\,L\,V_o\,\bar{e}_z} \quad (5)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- Adiabatic, workless flow keeps the stagnation enthalpy (2); the pressure drop to p_a lowers the density, so the gas accelerates ($V_o > V_u$) and cools slightly ($T_o < T_u$).
- Both legs push the pipe upwards: the gas turning down in the right leg and the gauge pressure at the measuring section. The moment comes only from the outlet jet, offset by L from O.
- Incompressible-like limit ($p_u \to p_a$, small V): $T_o \to T_u$, $V_o \to V_u$, $F_y \to 2\rho_u V_u^2 A$.
- Dimensions (SI): $[\rho_u V_u^2 A] = \mathrm{N}$; $[\rho_u V_u A L V_o] = \mathrm{N\cdot m}$; $[R_g T] = [V^2] = \mathrm{m^2/s^2}$.
- Numerical illustration (hypothetical data, air): $V_u = 50$ m/s, $p_u = 1.5\times10^5$ Pa, $T_u = 400$ K, $p_a = 101325$ Pa, $R_g = 287$ J/(kg K), $\gamma = 1.4$, $A = 0.01$ m^2, $L = 1$ m give $T_o = 398.537$ K, $V_o = 73.749$ m/s, $F_y = 567.6$ N, $M_z = 48.181$ N·m (residuals of (1), (2) below 10^{-8}).

- Agreement with the official solution: CL12.pdf obtains (1), (2), (4) and (5); it does not solve (1)-(2) explicitly.